

## Writing a Scope of Inference

Guidelines:

1. Keep your discussion of causal inference SEPARATE from inference to populations. In other words, do not refer to both within the same sentence. Initially start by writing your whole *Scope of Inference* in just two sentences (you can get more creative after you have nailed the two sentence approach).
  1. one sentence addressing whether causal inference is justified and why
  2. one sentence addressing whether any inference to a larger population is justified and why
2. Make sure you write the *Scope of Inference* specific to the language of the problem (*not* just generic statements). For example, DO NOT say “There was no randomization so cause and effect statements cannot be made.” Instead, for example, say “The lizards were not randomly assigned to diets, therefore we can not conclude that the fly diet caused the increase in average weight over the lizards on the tofu diet.” So, a general format for starting the two sentences described above is...
  1. Say *what* was (or was not) randomly assigned to *what*, therefore...
  2. Say *what* was (or was not) randomly selected from *what*, therefore...
3. There is no need to refer to the statistical results - you can write a *Scope of Inference* based on the study design without yet knowing the results.

Some examples from *The Statistical Sleuth* (not all follow these specific guidelines)

- Because students were randomly assigned to type of questionnaire, one may infer the difference in creativity scores was *caused* by the difference in motivational questionnaires. Because the subjects were not selected randomly from any population, extending this inference to any students beyond this group is speculative. This deficiency, however, is minor; the causal conclusion is strong even if it applies only to the recruited subjects.
- Although there is convincing evidence that the males, as a group, received larger starting salaries than the females, this analysis alone cannot address whether this difference is attributable to sex discrimination because sex cannot be randomly assigned to employee. The evidence is consistent with discrimination, but other possible explanations cannot be ruled out; for example, the males may have had more years of previous experience. Inference cannot extend beyond the employees from this bank during this time period because there was not random selection from a larger group of employees.